

Locked Phenomenological Modulations under Stress: v13 Pre-CMB Stress-Test Survey on DESI DR2

LVC v13 — Lagrangian Variable Cosmology, Version 13

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May 2026

Abstract

Building on v12, which reported a locked Gaussian modulation $T(z) = 1 + A \cdot \exp[-((z-z_c)/w)^2]$ with $z_c = 1/\sqrt{3} + 1/(3\pi)$ and $w = 2/(3\pi)$ preferred over Λ CDM by $\Delta\text{BIC} = -11.57$ on a 37-point combined DR2 dataset, we run a stress-test survey before any large-scale data analysis. The tests cover: peak-region BAO leave-one-out (LOO), per-point χ^2 decomposition, position-free form search, DR1/DR2 free-fit consistency, fiducial Ω_m sensitivity, $f\sigma_8$ joint, r_d -free fair comparison, BAO survey decomposition, DR1+DR2 merged fit, family with continuous n , H_0 prior shake, BAO covariance scaling, D_M vs D_H separation, bootstrap subsampling, BBN constraint ($r_d=147$ forced), and a Planck 2018+BAO N_{eff} prior. Two candidates emerge from a search for position-free forms: C1 = the v12 Gaussian lock and C2 = $T(z) = 1 + A \cdot (z/w)^2 / (1+(z/w)^4)$ with $w = 2/\pi$. We report numerical results without interpretation. The companion script `lvc_v13_reproduce.py` reproduces all numbers.

1. Setup

The dataset is identical to v12: 13 binned Pantheon+ supernovae, 7 DESI DR2 BAO measurements (one isotropic D_V at $z=0.295$ and six anisotropic D_M/r_d , D_H/r_d pairs at $z = 0.510, 0.706, 0.934, 1.321, 1.484, 2.330$), BOSS DR12 ($z=0.38$), eBOSS DR16 ($z = 0.698, 0.845, 1.480, 2.334$), Planck $\theta_* = 0.010409$, and SH0ES $H_0 = 73.04 \pm 1.04$. Total $N = 37$.

Two candidate forms are carried through all stress tests:

C1 (v12 family $n=1$): $T(z) = 1 + A \cdot \exp[-((z-z_c)/w)^2]$, with $z_c = 1/\sqrt{3} + 1/(3\pi) \approx 0.6834$, $w = 2/(3\pi) \approx 0.2122$ locked. Free parameters: H_0 , Ω_m , A . $k = 3$.

C2 (PowRat): $T(z) = 1 + A \cdot (z/w)^2 / (1+(z/w)^4)$, with $w = 2/\pi \approx 0.6366$ locked. The bump peaks at $z = w$ with maximum $A/2$. $T(0) = 1$ by construction; $T(\infty) \rightarrow 1$ as $1/z^2$. Free parameters: H_0 , Ω_m , A . $k = 3$.

C2 was selected from a search over seven position-free forms (R6-v12, R6-quartic, R6-cube, PowRat, LogGauss, double-zero, R7-tied) as the form with the smallest free-fit χ^2 on DR2. The lock $w = 2/\pi$ was identified as the natural-constant value within 1.5% of the free best-fit w .

2. Baseline DR2 fits

Model	k	χ^2	BIC	$\Delta\text{BIC vs } \Lambda\text{CDM}$
ΛCDM	2	29.676	36.898	—
C1 (v12 $n=1$)	3	14.498	25.331	-11.567
C2 (PowRat $2/\pi$)	3	15.602	26.435	-10.464

Table 1. Baseline geometric DR2 fits (37 points, r_d derived from θ_*).

3. Stress-test results

3.1 Peak-region BAO leave-one-out

The DR2 BAO pair indices 1 (LRG2, $z=0.706$), 2 (LRG3+ELG1, $z=0.934$), and 4 (QSO, $z=1.484$) lie in the peak region. We remove subsets and refit:

Removed	N	C1 ΔBIC	C2 ΔBIC
none (baseline)	37	−11.57	−10.46
LRG2 ($z=0.706$)	36	−4.87	−5.75
LRG3+ELG1 ($z=0.934$)	36	−10.20	−11.13
QSO ($z=1.484$)	36	−11.72	−10.66
Ly α ($z=2.330$) [non-peak]	36	−12.17	−9.80
LRG2 + LRG3+ELG1 + QSO	34	−2.11	−4.37
DESI LRG2 + eBOSS LRG	35	−3.96	−4.33

Table 2. Peak-region BAO LOO. N excludes removed points.

3.2 Per-point χ^2 decomposition

Decomposing the C1 vs ΛCDM $\Delta\chi^2 = +12.57$ by data point on baseline DR2:

Point	z	$\Lambda\text{CDM } \chi^2$	C1 χ^2	$\Delta\chi^2$
DESI LRG2	0.706	6.34	0.25	+6.09
DESI LRG1	0.510	4.85	1.71	+3.14
BOSS DR12	0.380	2.24	0.67	+1.57
eBOSS LRG	0.698	4.50	3.27	+1.23
DESI LRG3+ELG1	0.934	1.26	0.24	+1.02
Other 8 points	—	6.94	7.43	−0.49
BAO total	—	26.13	13.56	+12.57

Table 3. χ^2 contribution per BAO measurement. LRG2 alone contributes 48% of the total $\Delta\chi^2$. The two LRG measurements (DESI $z=0.706$, eBOSS $z=0.698$) differ by 1.36σ in D_M/r_d .

3.3 Position-free form search

Seven position-free forms ($k=4$ with H_0 , Ω_m , A , w free) were fit on baseline DR2:

Form	χ^2	$\Delta\text{BIC vs } \Lambda\text{CDM}$ ($k=4$)	best A	best w
PowRat $(z/w)^2/[1+(z/w)^4]$	15.59	−6.86	+0.0745	0.625
R6-quartic $(z/w)^2 \cdot \exp[-(z/w)^4]$	15.78	−6.67	+0.0982	0.778
R6-cube $(z/w)^3 \cdot \exp[-(z/w)^2]$	16.37	−6.08	+0.0948	0.518
Double-zero $z^2(z-w) \cdot \exp$	16.93	−5.52	+0.0678	0.321
Log-Gauss $z \cdot \exp[-(\ln(1+z)/w)^2]$	17.00	−5.45	+0.0701	0.749
R6-v12 $(z/w) \cdot \exp[-(z/w)^2]$	18.15	−4.30	+0.0749	0.899
R7-tied $z_c = w/\sqrt{2}$	18.80	−3.65	+0.0452	0.771

Table 4. Position-free form search. PowRat has lowest χ^2 . The PowRat free best-fit $w = 0.625$ lies within 2% of natural-constant candidates $\{e/4 = 0.680, 2/\pi = 0.637, 1/\sqrt{3} = 0.577, \ln 2 = 0.693\}$.

Locking w to natural-constant values ($k=3$) on PowRat:

Lock	w value	χ^2	$\Delta\text{BIC vs } \Lambda\text{CDM (k=3)}$
$2/\pi$	0.6366	15.60	-10.46
$e/4$	0.6796	15.81	-10.26
$1/\sqrt{3}$	0.5774	15.82	-10.25
$\ln 2$	0.6931	15.91	-10.15

Table 5. PowRat with various w locks. $C2 = w = 2/\pi$ is selected for the rest of the v13 stress survey.

3.4 DR1 vs DR2 free-fit consistency

Dataset	Form	χ^2	free w	free z_c	free A	ΔBIC
DR1	$\Lambda\text{CDM (k=2)}$	27.97	—	—	—	—
DR1	PowRat free (k=4)	20.23	0.580	—	+0.075	-0.58
DR1	Gaussian free (k=5)	16.95	0.106	0.583	+0.105	-0.27
DR2	$\Lambda\text{CDM (k=2)}$	29.68	—	—	—	—
DR2	PowRat free (k=4)	15.59	0.625	—	+0.074	-6.86
DR2	Gaussian free (k=5)	14.48	0.208	0.679	+0.049	-4.36

Table 6. Free-fit on DR1 and DR2. PowRat w shifts by 7.7% between DR1 and DR2; Gaussian z_c shifts by 16.5%.

3.5 Fiducial Ω_m sensitivity

Pantheon+ binned MU values were rebuilt at five fiducial Ω_m values and the DR2 fit repeated:

Ω_m fid	$\Lambda\text{CDM } \chi^2$	PowRat χ^2	PowR at w	PowRat ΔBIC	Gauss χ^2	Gauss z_c / w	Gauss ΔBIC
0.270	26.24	18.56	0.748	-0.46	14.96	0.703 / 0.179	-0.45
0.290	26.18	16.80	0.705	-2.16	13.79	0.696 / 0.186	-1.57
0.300	26.56	16.24	0.685	-3.10	13.57	0.692 / 0.190	-2.16
0.320	28.07	15.64	0.649	-5.21	13.82	0.685 / 0.200	-3.41
0.334	29.68	15.59	0.625	-6.86	14.48	0.679 / 0.208	-4.36

Table 7. Fiducial Ω_m sensitivity. Both forms weaken monotonically as Ω_m fid decreases.

3.6 $f\sigma_8$ joint analysis

Linear matter growth ODE solved with the modified background; 19 RSD $f\sigma_8$ measurements added to the geometric likelihood. $N = 37 + 19 = 56$.

Model	k	χ^2 total	χ^2 geom	$\chi^2 f\sigma_8$	BIC	$\Delta\text{BIC vs } \Lambda\text{CDM}$
ΛCDM	3	39.91	29.68	10.24	51.99	—
C1 (v12 n=1)	4	24.64	14.50	10.14	40.74	-11.25
C2 (PowRat $2/\pi$)	4	25.44	15.61	9.83	41.54	-10.45

Table 8. Joint geometry + RSD fit.

3.7 r_d -free fair comparison

Model	k	χ^2	r_d (Mpc)	$\Delta\text{BIC vs } \Lambda\text{CDM (k=3)}$
$\Lambda\text{CDM, } r_d\text{-free}$	3	27.17	139.59	—

Model	k	χ^2	r_d (Mpc)	$\Delta\text{BIC vs } \Lambda\text{CDM (k=3)}$
C1 + r_d -free	4	13.55	137.37	−10.02
C2 + r_d -free	4	15.40	136.35	−8.16

Table 9. r_d -free comparison. C1 retains very-strong evidence ($\Delta\text{BIC} \leq -10$); C2 falls to strong evidence.

3.8 BAO survey decomposition

Subset	N	$\Lambda\text{CDM } \chi^2$	C1 ΔBIC	C2 ΔBIC
SN + DESI only (no BOSS, eBOSS)	22	21.39	−8.85	−8.54
SN + BOSS + eBOSS only (no DESI)	24	7.86	−0.27	+0.11
SN + DESI + BOSS only (no eBOSS)	24	24.08	−10.73	−9.52

Table 10. BAO survey decomposition. The ΔBIC concentrates in DESI; BOSS+eBOSS alone yield $\Delta\text{BIC} \approx 0$.

3.9 DR1+DR2 merged fit

Model	k	χ^2	$\Delta\text{BIC vs } \Lambda\text{CDM (N=49)}$
ΛCDM	2	46.40	—
C1 (v12 n=1)	3	29.86	−12.65
C2 (PowRat 2/ π)	3	30.18	−12.33
PowRat $w=1/\sqrt{3}$	3	29.78	−12.73

Table 11. DR1+DR2 merged fit. $N = 49$ (13 SN + 12 DESI DR1 + 13 DESI DR2 + 2 BOSS + 7 eBOSS + θ_* + H_0). All locks fit jointly without inflation.

3.10 Family with continuous n (honest accounting)

Dataset	Lock χ^2 (k=3)	Free-n χ^2 (k=4)	Lock ΔBIC	Free-n ΔBIC	best n
DR2	14.498 (n=1)	14.484	−11.57	−7.97	0.96
DR1	17.006 (n=0)	16.959	−7.38	−3.84	0.03

Table 12. Family with n treated as continuous free parameter. The integer lock yields ΔBIC advantage of +3.6 over the continuous case in both DR1 and DR2.

3.11 H_0 prior shake

H_0 prior	$\Lambda\text{CDM } H_0$	C1 χ^2	C1 ΔBIC	C2 χ^2	C2 ΔBIC
SH0ES (73.04 ± 1.04)	73.04	14.50	−11.57	15.60	−10.46
TRGB-like (69.8 ± 1.7)	69.80	14.50	−11.57	15.60	−10.46
Planck-like (67.4 ± 0.5)	67.40	14.50	−11.57	15.60	−10.46
No H_0 prior	free	14.50	−11.59	15.60	−10.49

Table 13. H_0 prior shake. χ^2 and ΔBIC are independent of the H_0 prior choice; each model auto-anchors at the prior mean.

3.12 BAO covariance scaling

Covariance	$\Lambda\text{CDM } \chi^2$	C1 χ^2	C1 ΔBIC	C2 χ^2	C2 ΔBIC
corr $\times 0.5$	28.20	13.64	−10.94	15.38	−9.21
corr $\times 1.0$ (baseline)	29.68	14.50	−11.57	15.60	−10.46

Covariance	$\Lambda\text{CDM } \chi^2$	C1 χ^2	C1 ΔBIC	C2 χ^2	C2 ΔBIC
corr $\times 1.5$	35.63	17.87	-14.16	18.24	-13.79
corr = 0	29.03	14.04	-11.38	16.54	-8.87

Table 14. BAO covariance scaling. Setting all reported pair correlations to zero weakens C2's ΔBIC by 1.59; C1's ΔBIC weakens by 0.19.

3.13 D_M vs D_H separation

Subset	N	$\Lambda\text{CDM } \chi^2$	C1 ΔBIC	C2 ΔBIC
Both D_M and D_H (baseline)	37	29.68	-11.57	-10.46
D_M only (+ D_V)	27	13.96	-2.44	-2.83
D_H only (no D_V)	25	17.77	-8.93	-6.12

Table 15. D_M vs D_H separation. The bump signal concentrates in D_H : with D_M only, both models reach only positive evidence ($\Delta\text{BIC} \approx -2.5$).

3.14 Bootstrap subsample

40 random subsets of effective $N \approx 24$ (always including SN block, θ_* , and H_0 prior). For each, refit ΛCDM , C1, C2:

Candidate	A median	A IQR	ΔBIC median	ΔBIC IQR	wins $\Delta\text{BIC} < 0$	wins < -5
C1 (v12 n=1)	+0.0491	[+0.046, +0.056]	-7.80	[-9.35, -5.04]	39/40	30/40
C2 (PowRat 2/ π)	+0.0749	[+0.067, +0.086]	-7.11	[-8.72, -5.60]	40/40	33/40

Table 16. Bootstrap (40 iterations). C1 wins (lowest ΔBIC) in 21/40 = 52.5%, C2 in 16/40 = 40%, third candidate (PowRat 1/ $\sqrt{3}$) in 3/40 = 7.5%.

3.15 BBN constraint ($r_d = 147.05$ forced)

Each candidate's preferred r_d implies an N_{eff} via $r_d \approx 147.05 \cdot (3.046/N_{\text{eff}})^{0.246}$. Forcing $r_d = 147.05$ (BBN-compatible value):

Model	χ^2 ($r_d=147$)	$\Delta\text{BIC vs } \Lambda\text{CDM}$ ($r_d=147$)	baseline ΔBIC	strength loss
ΛCDM	38.52	—	—	—
C1 (v12 n=1)	31.42	-3.49	-11.57	+8.08
C2 (PowRat 2/ π)	35.26	+0.35	-10.46	+10.82

Table 17. BBN constraint via $r_d=147$ lock. C1 keeps positive evidence; C2's ΔBIC becomes positive (worse than ΛCDM).

3.16 Planck N_{eff} prior

Adding the Planck 2018+BAO measurement $N_{\text{eff}} = 2.99 \pm 0.17$ (Planck Collaboration, A&A 641, A6, 2020) as one Gaussian likelihood term using $r_d \approx 147.05 \cdot (3.046/N_{\text{eff}})^{0.246}$. $N = 38$.

Model	χ^2_{total}	χ^2_{geom}	$\chi^2(N_{\text{eff}})$	r_d	N_{eff}	$\Delta\text{BIC vs } \Lambda\text{CDM}$
ΛCDM	36.71	33.84	2.86	144.42	3.28	—
C1 (v12 n=1)	26.47	22.00	4.47	143.66	3.35	-6.60
C2 (PowRat 2/ π)	29.73	24.84	4.89	143.48	3.37	-3.34

Table 18. Planck N_{eff} prior added. C1 retains strong evidence ($\Delta\text{BIC} = -6.60$); C2 falls to positive evidence (-3.34).

4. Summary across all stress tests

Test	C1 (v12 n=1)	C2 (PowRat 2/n)
Baseline DR2 ΔBIC	-11.57	-10.46
Worst single-BAO LOO ΔBIC	-4.87 (LRG2)	-5.75 (LRG2)
Peak 3-pt LOO ΔBIC	-2.11	-4.37
DR1/DR2 free param shift	z_c : 16.5%	w: 7.7%
Fiducial $\Omega_m = 0.27$ ΔBIC	-0.45	-0.46
σ_8 joint ΔBIC	-11.25	-10.45
r_d -free ΔBIC	-10.02	-8.16
DESI-only ΔBIC	-8.85	-8.54
BOSS+eBOSS only ΔBIC	-0.27	+0.11
DR1+DR2 merged ΔBIC	-12.65	-12.33
BAO corr=0 ΔBIC	-11.38	-8.87
D_H only ΔBIC	-8.93	-6.12
Bootstrap win rate	52.5%	40.0%
BBN forced ($r_d=147$) ΔBIC	-3.49	+0.35
Planck N_{eff} prior ΔBIC	-6.60	-3.34

Table 19. Summary across all 16 stress tests. Numerical results without interpretation.

5. Reproducibility

All numerical results above are reproduced by `lvc_v13_reproduce.py`. The script extends `lvc_v12_reproduce.py` with the additional stress tests (§3.1–§3.16). Tabulated χ^2 values reproduce within ≤ 0.05 and BIC values within ≤ 0.10 . Computational platform: Python 3, `scipy differential_evolution` + Nelder-Mead (multi-seed).

External datum used in §3.16: Planck 2018 + BAO $N_{\text{eff}} = 2.99 \pm 0.17$, taken from Planck Collaboration, A&A 641, A6 (2020), Table 2 (extended ΛCDM , parameter N_{eff}). The r_d - N_{eff} scaling formula used is the Aylor et al. (2019) approximation, accurate within 0.5% of full Boltzmann recombination calculations near the standard cosmology.

6. Note on scope

This v13 paper reports stress-test results only. No theoretical interpretation, no claim of mechanism, no commitment to either C1 or C2 as the preferred form. The companion working note (V13_WORKNOTE.txt, internal) records additional context not included here.

External-data tests beyond §3.15 and §3.16 (Pantheon+ unbinned, full Planck likelihood, lensing A_L , DESI Year 3) are not included in this version and are deferred to future work.